

#question 1

```
import pandas as pd
data = {
    "ID": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    "Refund": ["Yes", "No", "No", "Yes", "No", "No", "Yes", "No", "No", "No"]
    "Marital Status": ["Single", "Married", "Single", "Married", "Divorced",
        "Married", "Divorced", "Single", "Married", "Single"]
    "Taxable Income": ["125K", "100K", "70K", "120K", "95K",
        "60K", "220K", "85K", "75K", "90K"],
    "Cheat": ["No", "No", "No", "No", "Yes", "No", "No", "Yes", "No", "No"]
}
df=pd.DataFrame(data)
print(df)
```

	ID	Refund	Marital Status	Taxable Income	Cheat
0	1	Yes	Single	125K	No
1	2	No	Married	100K	No
2	3	No	Single	70K	No
3	4	Yes	Married	120K	No
4	5	No	Divorced	95K	Yes
5	6	No	Married	60K	No
6	7	Yes	Divorced	220K	No
7	8	No	Single	85K	Yes
8	9	No	Married	75K	No
9	10	No	Single	90K	No

#question 2

```
df.iloc[[0,4,7,8]]
```

	ID	Refund	Marital Status	Taxable Income	Cheat
0	1	Yes	Single	125K	No
4	5	No	Divorced	95K	Yes
7	8	No	Single	85K	Yes
8	9	No	Married	75K	No

#question 3

```
df.loc[3:7]
```

	ID	Refund	Marital Status	Taxable Income	Cheat	
3	4	Yes	Married	120K	No	
4	5	No	Divorced	95K	Yes	
5	6	No	Married	60K	No	
6	7	Yes	Divorced	220K	No	
7	8	No	Single	85K	Yes	

```
df.iloc[4:9, 2:5]
```

	Marital Status	Taxable Income	Cheat	
4	Divorced	95K	Yes	
5	Married	60K	No	
6	Divorced	220K	No	
7	Single	85K	Yes	
8	Married	75K	No	

```
df.iloc[:, 1:4]
```

	Refund	Marital Status	Taxable Income	
0	Yes	Single	125K	
1	No	Married	100K	
2	No	Single	70K	
3	Yes	Married	120K	
4	No	Divorced	95K	
5	No	Married	60K	
6	Yes	Divorced	220K	
7	No	Single	85K	
8	No	Married	75K	
9	No	Single	90K	

```
#question 4
df = pd.read_csv("Iris.csv")
```

```
print(df.head())
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa

1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

#question 5

```
df_modified = df.drop(index=4, columns=df.columns[3])
```

```
print(df_modified)
```

	Id	SepalLengthCm	SepalWidthCm	PetalWidthCm	Species
0	1	5.1	3.5	0.2	Iris-setosa
1	2	4.9	3.0	0.2	Iris-setosa
2	3	4.7	3.2	0.2	Iris-setosa
3	4	4.6	3.1	0.2	Iris-setosa
5	6	5.4	3.9	0.4	Iris-setosa
..	...	...	...	...	...
145	146	6.7	3.0	2.3	Iris-virginica
146	147	6.3	2.5	1.9	Iris-virginica
147	148	6.5	3.0	2.0	Iris-virginica
148	149	6.2	3.4	2.3	Iris-virginica
149	150	5.9	3.0	1.8	Iris-virginica

[149 rows x 5 columns]

#question 6

```
import pandas as pd
```

```
data = {
    'Employee_ID': [101, 102, 103, 104, 105],
    'Name': ['Alice', 'Bob', 'Charlie', 'Diana', 'Edward'],
    'Department': ['HR', 'IT', 'IT', 'Marketing', 'Sales'],
    'Age': [29, 34, 41, 28, 38],
    'Salary': [50000, 70000, 65000, 55000, 60000],
    'Years_of_Experience': [4, 8, 10, 3, 12],
    'Joining_Date': ['2020-03-15', '2017-07-19', '2013-06-01', '2021-02-10',
    'Gender': ['Female', 'Male', 'Male', 'Female', 'Male'],
    'Bonus': [5000, 7000, 6000, 4500, 5000],
    'Rating': [4.5, 4.0, 3.8, 4.7, 3.5]
}
```

```
df = pd.DataFrame(data)
df.to_csv('employees.csv', index=False)
df = pd.read_csv('employees.csv')
```

```
print("no of rows and columns", df.shape)
```

no of rows and columns (5, 10)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Employee_ID           5 non-null     int64
1   Name                  5 non-null     object
2   Department            5 non-null     object
3   Age                   5 non-null     int64
4   Salary                5 non-null     int64
5   Years_of_Experience   5 non-null     int64
6   Joining_Date          5 non-null     object
7   Gender                5 non-null     object
8   Bonus                5 non-null     int64
9   Rating                5 non-null     float64
dtypes: float64(1), int64(5), object(4)
memory usage: 532.0+ bytes
```

```
print(df.describe())
```

```

count      Employee_ID      Age      Salary      Years_of_Experience      Bonus  \
count      5.000000      5.000000      5.00000      5.000000      5.0
mean      103.000000      34.000000      60000.00000      7.400000      5500.0
std        1.581139      5.612486      7905.69415      3.847077      1000.0
min        101.000000      28.000000      50000.00000      3.000000      4500.0
25%        102.000000      29.000000      55000.00000      4.000000      5000.0
50%        103.000000      34.000000      60000.00000      8.000000      5000.0
75%        104.000000      38.000000      65000.00000      10.000000     6000.0
max        105.000000      41.000000      70000.00000      12.000000     7000.0

count      Rating
count      5.000000
mean      4.100000
std        0.494975
min        3.500000
25%        3.800000
50%        4.000000
75%        4.500000
max        4.700000
```

```
print("First 5 rows:")
print(df.head(5))
print("Last 3 rows:")
print(df.tail(3))
```

```
First 5 rows:
Employee_ID      Name      Department      Age      Salary      Years_of_Experience  \
0           101      Alice              HR      29      50000              4
1           102        Bob              IT      34      70000              8
2           103     Charlie              IT      41      65000             10
3           104      Diana      Marketing      28      55000              3
4           105     Edward              Sales      38      60000             12

Joining_Date      Gender      Bonus      Rating
0      2020-03-15     Female      5000      4.5
1      2017-07-19      Male      7000      4.0
2      2013-06-01      Male      6000      3.8
3      2021-02-10     Female      4500      4.7
```

```

4   2010-11-25   Male   5000   3.5
Last 3 rows:
   Employee_ID   Name Department   Age   Salary   Years_of_Experience \
2           103  Charlie         IT   41   65000           10
3           104   Diana  Marketing   28   55000           3
4           105   Edward     Sales   38   60000          12

   Joining_Date   Gender   Bonus   Rating
2   2013-06-01    Male    6000    3.8
3   2021-02-10  Female    4500    4.7
4   2010-11-25    Male    5000    3.5

```

```

avg_salary = df["Salary"].mean()
total_bonus = df["Bonus"].sum()
youngest_age = df["Age"].min()
highest_rating = df["Rating"].max()

```

```

print(avg_salary)
print(total_bonus)
print(youngest_age)
print(highest_rating)

```

```

60000.0
27500
28
4.7

```

```

df_sorted = df.sort_values(by="Salary", ascending=False)
print(df_sorted)

```

```

   Employee_ID   Name Department   Age   Salary   Years_of_Experience \
1           102    Bob         IT   34   70000           8
2           103  Charlie         IT   41   65000          10
4           105   Edward     Sales   38   60000          12
3           104   Diana  Marketing   28   55000           3
0           101   Alice         HR   29   50000           4

   Joining_Date   Gender   Bonus   Rating
1   2017-07-19    Male    7000    4.0
2   2013-06-01    Male    6000    3.8
4   2010-11-25    Male    5000    3.5
3   2021-02-10  Female    4500    4.7
0   2020-03-15  Female    5000    4.5

```

```

def categorize_rating(rating):
    if rating >= 4.5:
        return "Excellent"
    elif rating >= 4.0:
        return "Good"
    else:
        return "Average"

df["Performance_Category"] = df["Rating"].apply(categorize_rating)
print(df[["Name", "Rating", "Performance_Category"]])

```

	Name	Rating	Performance_Category
0	Alice	4.5	Excellent
1	Bob	4.0	Good
2	Charlie	3.8	Average
3	Diana	4.7	Excellent
4	Edward	3.5	Average

```
print("Missing values count:")
print(df.isnull().sum())
```

```
Missing values count:
Employee_ID      0
Name              0
Department        0
Age               0
Salary            0
Years_of_Experience  0
Joining_Date      0
Gender            0
Bonus             0
Rating            0
Performance_Category  0
dtype: int64
```

```
df.rename(columns={"Employee_ID": "ID"}, inplace=True)
print(df.columns)
```

```
Index(['ID', 'Name', 'Department', 'Age', 'Salary', 'Years_of_Experience',
      'Joining_Date', 'Gender', 'Bonus', 'Rating', 'Performance_Category'],
      dtype='object')
```

```
exp_5 = df[df["Years_of_Experience"] > 5]
print("Employees with > 5 years exp")
print(exp_5[["Name", "Years_of_Experience"]])
```

```
Employees with > 5 years exp
      Name  Years_of_Experience
1      Bob                   8
2  Charlie                  10
4  Edward                   12
```

```
it_dept = df[df["Department"] == "IT"]
print("Employees in IT department:")
print(it_dept[["Name", "Department"]])
```

```
Employees in IT department:
      Name Department
1      Bob         IT
2  Charlie         IT
```

```
df["Tax"] = df["Salary"] * 0.10
print(df[["Name", "Salary", "Tax"]])
```

```
      Name  Salary    Tax
0  Alice   50000  5000.0
```

1	Bob	70000	7000.0
2	Charlie	65000	6500.0
3	Diana	55000	5500.0
4	Edward	60000	6000.0

```
df.to_csv("modified_employees.csv", index=False)  
print("File saved as modified_employees.csv successfully.")
```

File saved as modified_employees.csv successfully.

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